Exercise Sheet 10
Predicate Logic – Equivalences & Semantics

---

1. The goal of this question is to prove the following formula in constructive Natural Deduction: $(\forall x.(p(x) \lor \exists y.q(x,y))) \leftrightarrow (\forall x.\exists y.(p(x) \lor q(x,y)))$
   (a) First prove the left-to-right implication $(\forall x.(p(x) \lor \exists y.q(x,y))) \rightarrow (\forall x.\exists y.(p(x) \lor q(x,y)))$
   (b) Now prove the right-to-left implication $(\forall x.\exists y.(p(x) \lor q(x,y))) \rightarrow (\forall x.(p(x) \lor \exists y.q(x,y)))$
2. Using Natural Deduction proofs, prove the equivalence between the following two formulas: $\forall x.\forall y.q(x,y) \rightarrow p(x)$ and $\forall x.(\exists y.q(x,y)) \rightarrow p(x)$
3. Using Natural Deduction proofs, prove the equivalence between the following two formulas: $(\exists x.p(x)) \land (\exists y.q(y))$ and $\exists x.\exists y.p(x) \land q(y)$.
4. Consider the signature that does not contain any function symbols, and that only contains the two unary predicate symbols p and q. Using the semantical approach, prove that $(\forall x.p(x) \land q(x)) \rightarrow \forall x.p(x)$.
5. Consider the following signature:
   - function symbols: `zero` of arity 0 and `succ` of arity 1
   - predicate symbols: `even` of arity 1 and `odd` of arity 1

   Provide:
   - a model M_1 such that $\models_{M_1} \exists x.\mathtt{odd}(x) \land \neg\mathtt{even}(x)$
   - a model M_2 such that $\models_{M_2} \neg(\exists x.\mathtt{odd}(x) \land \neg\mathtt{even}(x))$